we would have a short circuit that would cause wires to melt, things to smoke, and other bad things to happen.

That is why it is important to remember two general rules about working with electronic circuits. The first is that electricity will follow the path of least resistance to ground. Accidently place a metal object or extraneous wire across any bare wires in our example circuit and rather than flowing through the lamp, the electrical current will jump through the metal thing instead creating a short circuit. This is also why **insulators**, materials that do not conduct electricity like plastic and rubber, are used to cover our wires.

The second thing to remember is that the available amount of energy in a circuit must all be used. This is where the smoke and bad things come in because we need to have a power source that matches the load in our circuit. If we accidentally hook the lamp up to 24 volts instead of providing it with a comfortable 9 volts, the lamp will get very bright and very hot for a very short amount of time. And then we won't have a complete circuit because the lamp will have moved on to the big place in the sky for unfortunate components.

Electricity

To make our circuit work we need to have electricity. Electrical current moves in a circle through a circuit from the point of highest potential to the lowest. For the purpose of discussion, in our circuit the current flows from the positive terminal of the battery, through the closed switch, into the lamp to produce light, and is completed when the used up current reaches the negative terminal of the battery. If we were to open the switch by turning it off, we would break the circuit and the electricity will not flow. This kind of flow is an example of **Direct Current** or DC, where the current flows though a circuit in one direction. DC is common in batteries and those black, little power transformers filling our kitchen drawers. The second way that electricity can flow through a circuit is by **Alternating Current** or AC. AC reverses polarity at regular intervals, 60 times per second in the US, and is the type of electrical current that comes into our homes and is available from our wall sockets. Figure 12-2 shows a representation of these two different currents.

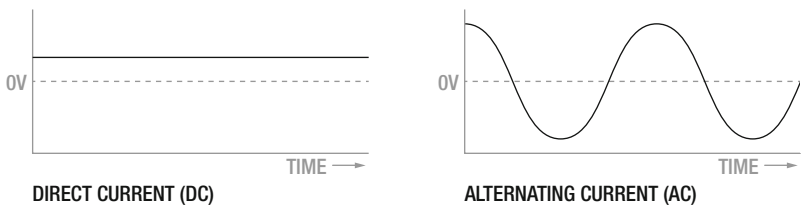


Figure 12-2. *Electrical currents*

There are three characteristics about our electrical circuit that are dependent on one another. The first of these is **current** or the amount of electrical energy that flows through a certain point in our circuit and is measured in **amperes** or **amps**. The second characteristic is **voltage** or the difference in potential between two points in a circuit, sometimes simply referred to as the circuit's electrical energy and is measured in **volts**. The third characteristic is the **resistance** placed on our circuit by our load is measured in **ohms**. The relationship between current, voltage, and resistance has been expressed in **Ohm's Law**, which can be summarized in the following equation:

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Voltage = Current × Resistance
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